Solutions – Quiz 1

ECON 308 – International Economic Relations

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1. **Data recap.** Unit labor requirements:

	Wine	Cheese
Home	$a_{LW} = 2$	$a_{LC} = 1$
Foreign	$a^*_{LW} = 6$	$a^*_{LC} = 2$

Labor: $L = L^* = 120$.

(a) **PPF (maximum production possibilities).**

- *Home:* With $L = 120$, the maximum Wine is $W^{\max} = 120/2 = 60$, and the maximum Cheese is $C^{\max} = 120/1 = 120$. The PPF is

$$2W + 1 \cdot C = 120 \quad \Longleftrightarrow \quad C = 120 - 2W.$$

- *Foreign:* With $L^* = 120$, the maximum Wine is $W^{\max *} = 120/6 = 20$, and the maximum Cheese is $C^{\max *} = 120/2 = 60$. The PPF is

$$6W + 2 \cdot C = 120 \quad \Longleftrightarrow \quad C = 60 - 3W.$$

(b) **Opportunity cost of Wine (in units of Cheese).** In the Ricardian model, the opportunity cost of Wine in terms of Cheese equals the ratio of unit labor requirements:

$$\mathrm{OC}_W^{\text{Home}} = \frac{a_{LW}}{a_{LC}} = \frac{2}{1} = 2 \text{ (Cheese per Wine)}, \qquad \mathrm{OC}_W^{\text{Foreign}} = \frac{a^*_{LW}}{a^*_{LC}} = \frac{6}{2} = 3.$$

Equivalently, the opportunity cost of Cheese in terms of Wine is the inverse:

$$\mathrm{OC}_C^{\text{Home}} = \frac{1}{2}, \qquad \mathrm{OC}_C^{\text{Foreign}} = \frac{1}{3}.$$

(c) **Comparative advantage.** A country has comparative advantage in the good with the lower opportunity cost:

$$\mathrm{OC}_W^{\text{Home}} = 2 < 3 = \mathrm{OC}_W^{\text{Foreign}} \Rightarrow \text{Home has C.A. in Wine.}$$

$$\mathrm{OC}_C^{\text{Foreign}} = \tfrac{1}{3} < \tfrac{1}{2} = \mathrm{OC}_C^{\text{Home}} \Rightarrow \text{Foreign has C.A. in Cheese.}$$

(d) **Trade pattern at world relative price $P_W/P_C = 1.5$.** In a one-factor Ricardian model, a country produces the good for which the world relative price exceeds its autarky unit-cost ratio (its opportunity cost). Home will produce/export Wine if

$$\frac{P_W}{P_C} > \frac{a_{LW}}{a_{LC}} = 2.$$

But $1.5 < 2$, so Home does *not* produce Wine; it produces Cheese.
Foreign will produce/export Wine if

$$\frac{P_W}{P_C} > \frac{a^*_{LW}}{a^*_{LC}} = 3.$$

But $1.5 < 3$, so Foreign also produces Cheese.
Conclusion: At $P_W/P_C = 1.5$, Wine is relatively cheap compared to both countries' opportunity costs, so *both* countries supply Cheese and import Wine (from the rest of the world). If we restrict the world to only these two countries, such a price cannot be an equilibrium; an equilibrium relative price must lie between 2 and 3 for mutually consistent specialization (Home in Wine, Foreign in Cheese).

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2. **Gravity equation:**

$$T_{ij} = A \cdot \frac{Y_i Y_j}{D_{ij}}, \quad A = 1.$$

Data: $Y_A = 2000,\ Y_B = 1500,\ Y_C = 500$ (all in billions), $D_{AB} = 1000$ km, $D_{AC} = 5000$ km.

(a) **Predicted trade T_{AB} and T_{AC}.**

$$T_{AB} = \frac{(2000)(1500)}{1000} = \frac{3{,}000{,}000}{1000} = 3000.$$

$$T_{AC} = \frac{(2000)(500)}{5000} = \frac{1{,}000{,}000}{5000} = 200.$$

$$\boxed{T_{AB} = 3000, \quad T_{AC} = 200 \quad \text{(in proportional units).}}$$

(b) **Trade ratio T_{AB}/T_{AC}.**

$$\frac{T_{AB}}{T_{AC}} = \frac{3000}{200} = 15.$$

$$\boxed{T_{AB} \text{ is 15 times } T_{AC}.}$$

(c) **FTA halves effective distance for A–C:** $D'_{AC} = 2500$ km. Recompute:

$$T'_{AC} = \frac{(2000)(500)}{2500} = \frac{1{,}000{,}000}{2500} = 400.$$

$$\boxed{T'_{AC} = 400.}$$

Effect: A halving of effective distance doubles predicted trade (from 200 to 400). The A–B vs. A–C ratio falls from 15 to $3000/400 = 7.5$, indicating a substantial relative increase in A–C trade intensity.